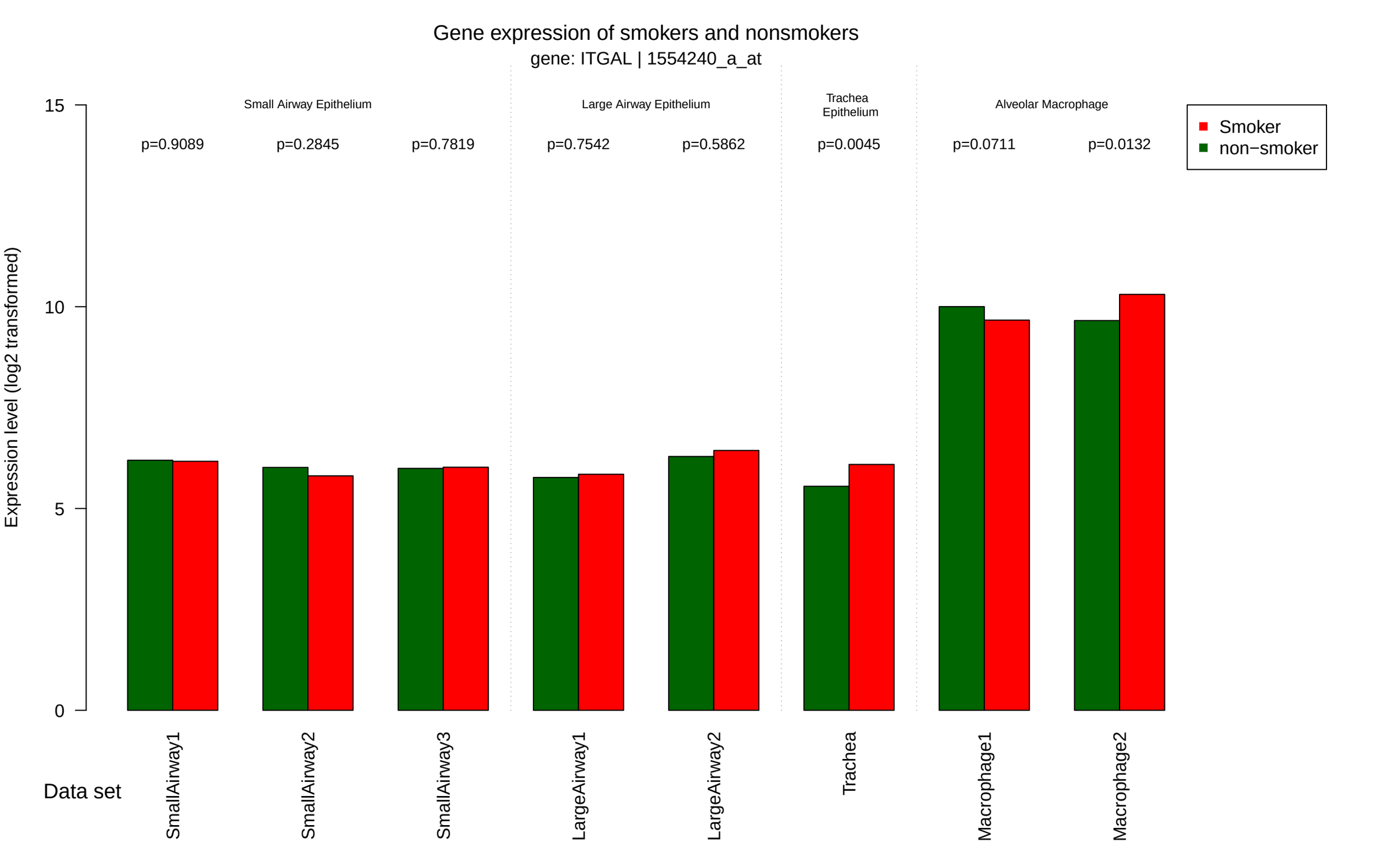
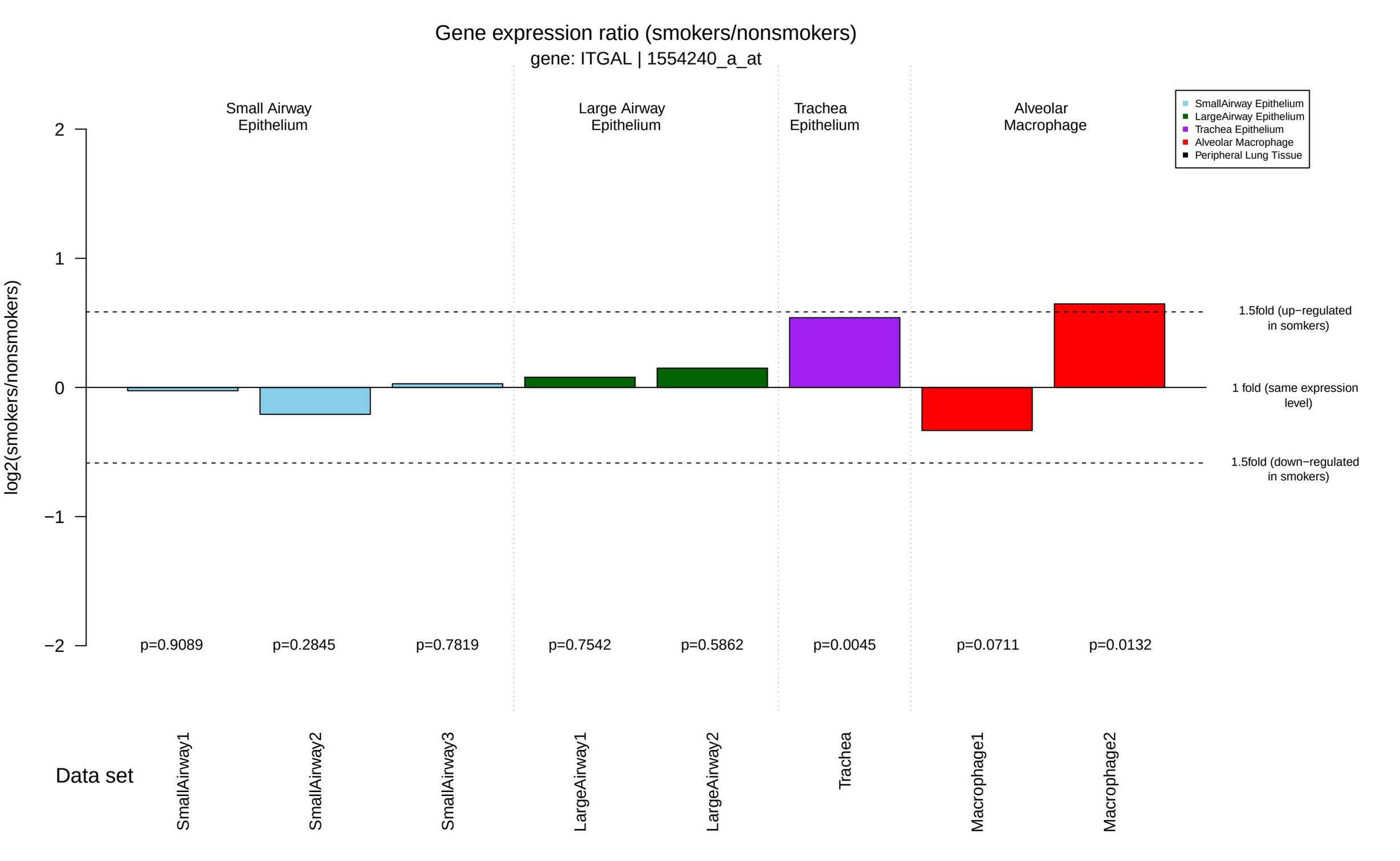
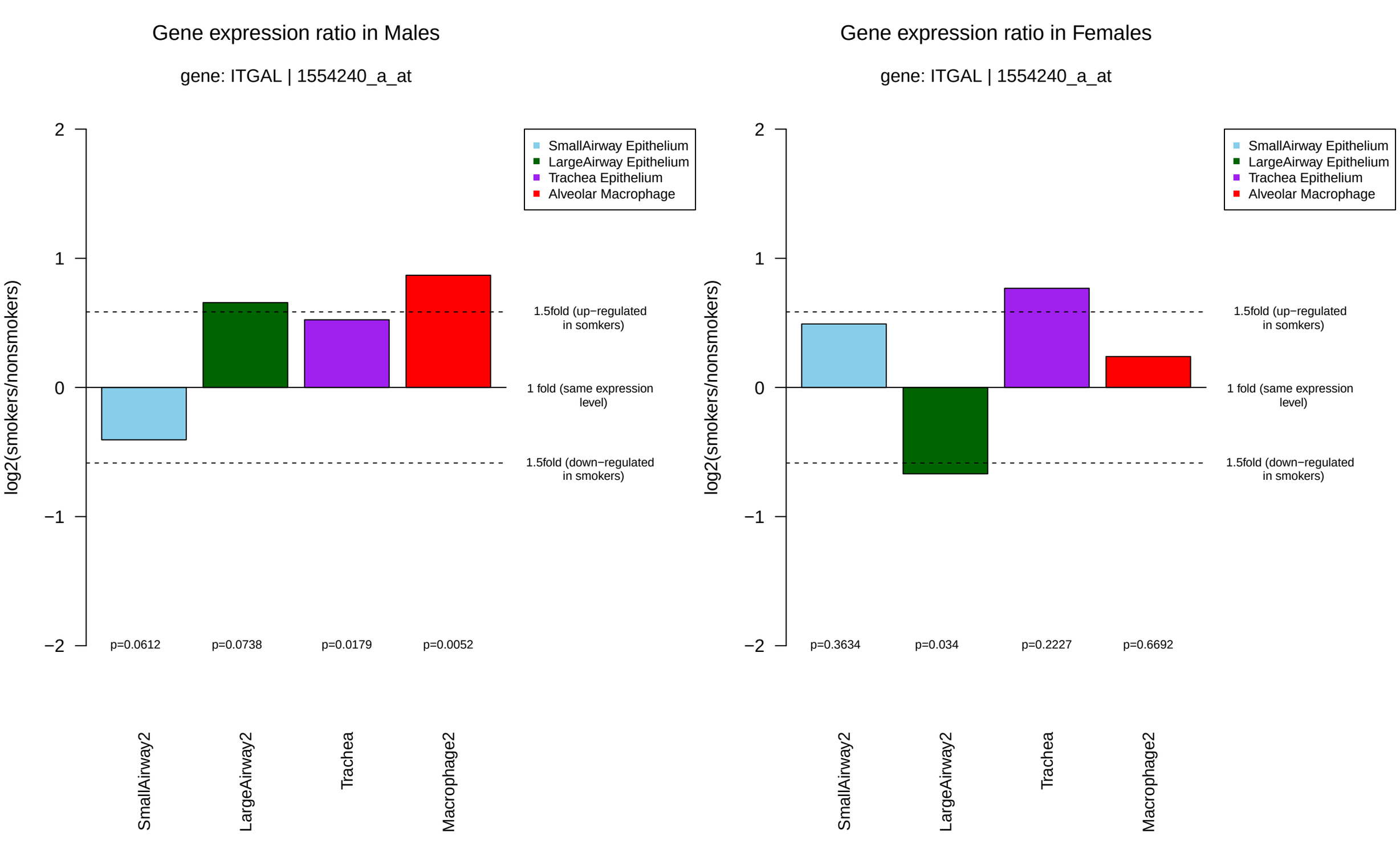
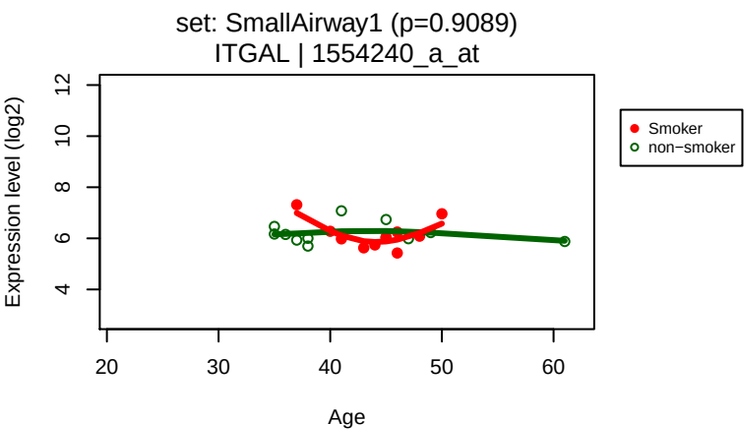
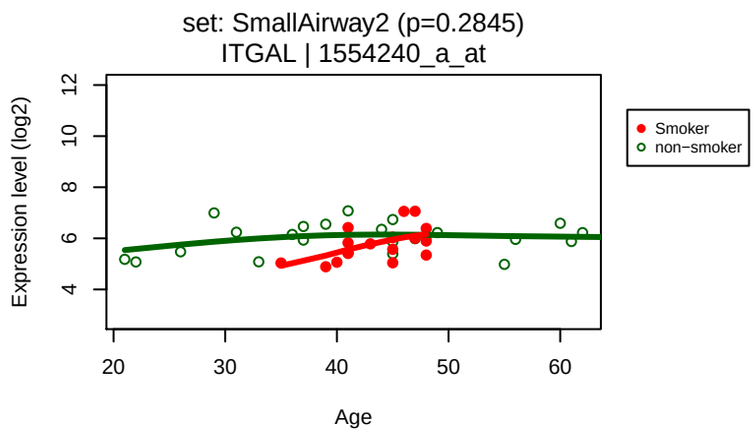
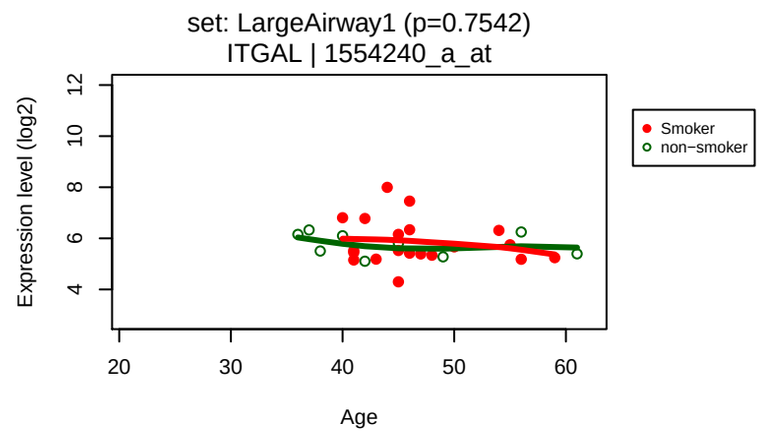

Gene expression of smokers and nonsmokers
gene: ITGAL | 1554240_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9089
p=0.2845
p=0.7819
p=0.7542
p=0.5862
p=0.0045
p=0.0711
p=0.0132
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

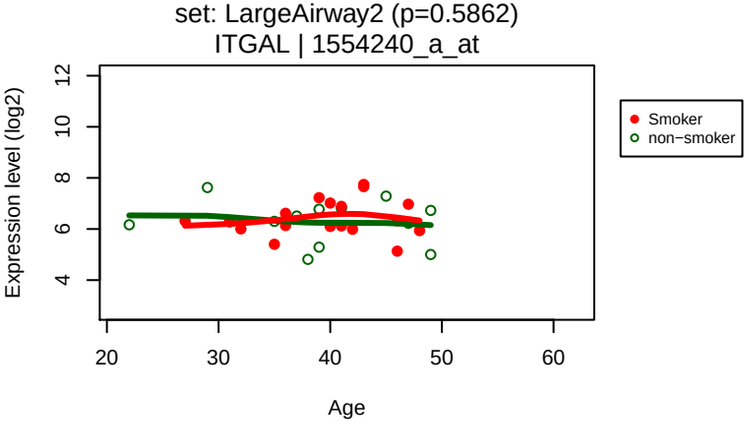

Gene expression ratio (smokers/nonsmokers)
gene: ITGAL | 1554240_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9089
p=0.2845
p=0.7819
p=0.7542
p=0.5862
p=0.0045
p=0.0711
p=0.0132
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

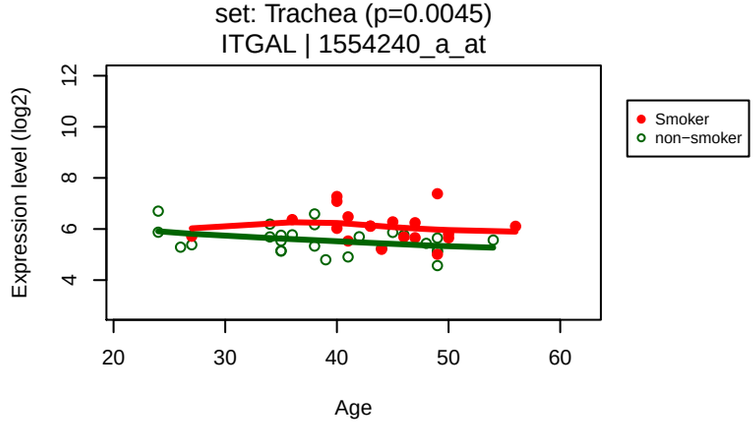

Gene expression ratio in Males
gene: ITGAL | 1554240_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0612
p=0.0738
p=0.0179
p=0.0052
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ITGAL | 1554240_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3634
p=0.034
p=0.2227
p=0.6692
SmallAirway2
LargeAirway2
Trachea
Macrophage2

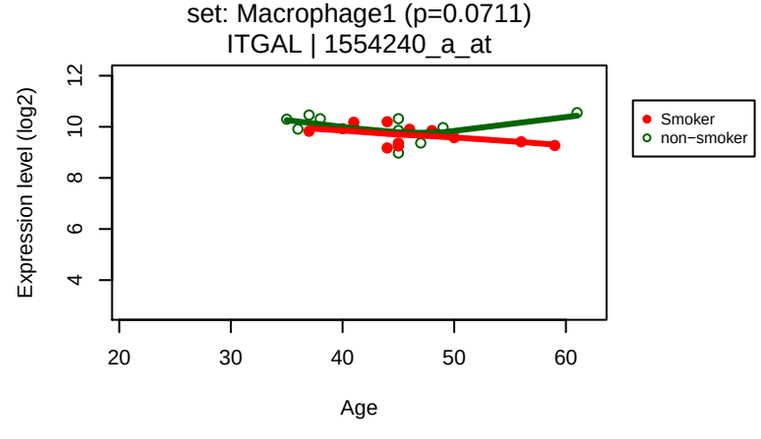

set: SmallAirway1 (p=0.9089)
ITGAL | 1554240_a_at
Expression level (log2)
Age
Smoker
non−smoker

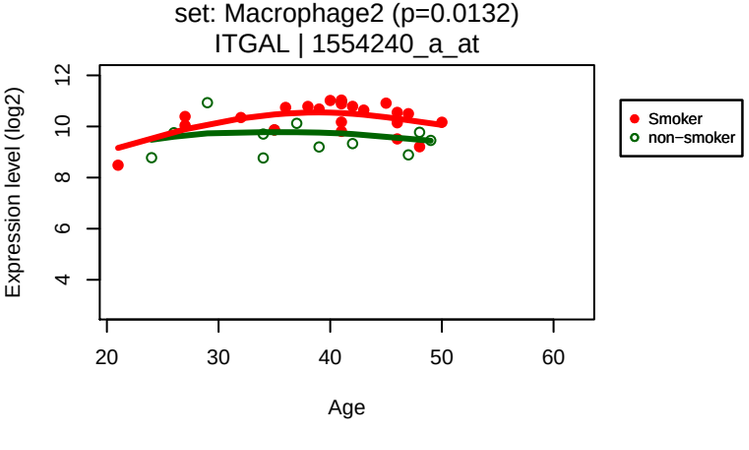

set: SmallAirway2 (p=0.2845)
ITGAL | 1554240_a_at
Expression level (log2)
Age
Smoker
non−smoker

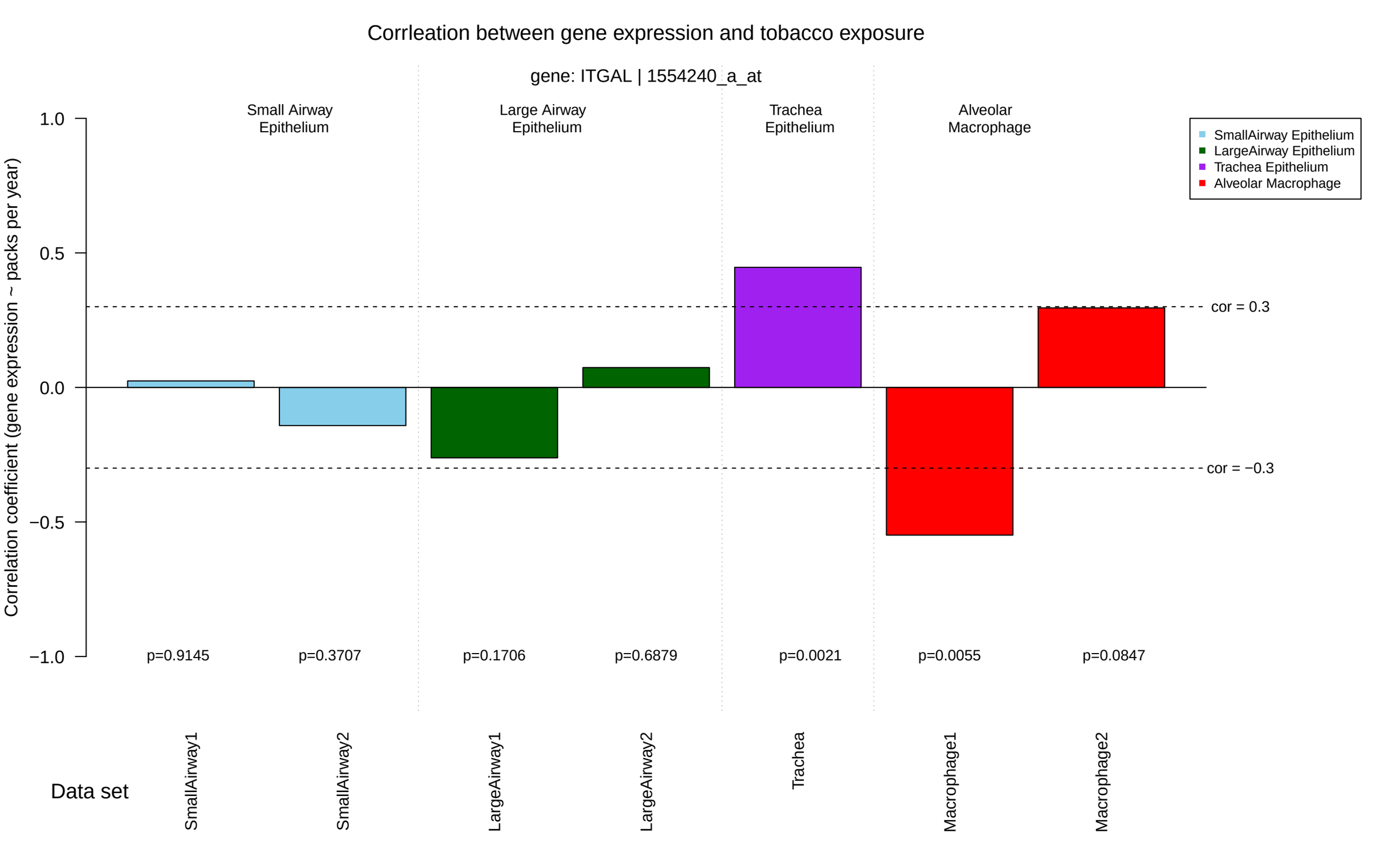

set: LargeAirway1 (p=0.7542)
ITGAL | 1554240_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5862)
ITGAL | 1554240_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0045)
ITGAL | 1554240_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0711)
ITGAL | 1554240_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0132)
ITGAL | 1554240_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ITGAL | 1554240_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.9145
p=0.3707
p=0.1706
p=0.6879
p=0.0021
p=0.0055
p=0.0847
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2